

Ashwin S

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Summary

Computer Science Engineering student with hands-on experience in Machine Learning, NLP, Data Analytics, and full-stack application development. Experienced in building practical AI-driven systems using Python, FastAPI, React, SQL, and modern ML frameworks, with proven success in hackathons and research publication. Strong interest in developing scalable, real-world solutions at the intersection of AI and software engineering.

Education

Sri Ramakrishna Engineering College, Coimbatore B.E. Computer Science and Engineering (Anna University) CGPA: 8.15 / 10	2024 - 2028 Coimbatore, IN
RJ Matriculation Higher Secondary School, Coimbatore HSC — Computer Science Stream (2024) SSLC (2022)	2022 - 2024

Technical Skills

- **Languages:** Java, C, C++, Python
- **AI / ML & Data:** Machine Learning, NLP (Transformers), Agentic AI, CatBoost, Pandas, NumPy, Matplotlib, Predictive & Geospatial Analytics
- **Web & Backend:** FastAPI, React.js, SQLAlchemy, REST APIs
- **Databases:** MySQL, PostgreSQL
- **Tools & Platforms:** Git, GitHub, VS Code, ServiceNow AI, Celonis Process Intelligence

Projects

Workforce Contribution Monitor

January 2026

- Built a full-stack monitoring system (**React + FastAPI + SQLAlchemy**) that scores employees on task completion and work impact instead of online presence, to reduce bias in performance reviews.
- Automated multi-source ingestion and normalization of communication- and execution-platform data through **real-time pipelines** on a modular, scalable backend.
- Designed a **custom scoring algorithm** separating high-activity from high-impact contributions, surfaced through an interactive React dashboard.

Tech Stack: Python, React.js, FastAPI, SQLAlchemy, SQL, Machine Learning, Data Analytics

Satellite-Based Water Quality Monitoring & Prediction

- Developed a **CatBoost Regressor** pipeline to predict water quality parameters from satellite imagery and geospatial data, with features engineered from spectral and environmental signals.
- Automated data ingestion, preprocessing and visualization for spatial and temporal trend analysis, enabling scalable estimation without manual field sampling.

Tech Stack: Python, CatBoost, Pandas, NumPy, Matplotlib, Geospatial Analytics

AI Customer Feedback Intelligence & Retention System

March 2026

- Built a multi-channel feedback system using **Transformer-based NLP** for sentiment analysis and behavioral modeling to flag high-risk customers before they churn.
- Automated a personalized retention pipeline (**FastAPI + PostgreSQL**) triggering targeted email and WhatsApp follow-ups scaled to each customer’s churn risk.
- Integrated a **multi-agent decision framework** that recommends refunds, escalations or retention offers to cut preventable revenue loss.

Tech Stack: Python, NLP (Transformers), FastAPI, React, PostgreSQL, WhatsApp API

Publications

[A High-Resolution Data Fusion Framework for Water Body Assessment Using Sentinel-2 and Gradient Boosting](#)
[IEEE Xplore](#) | [DOI](#) | ICCPCT 2026 | IEEE Kerala | 2026

Hackathons & Achievements

- **First Prize** — HackTIDE 24-Hour Hackathon, Amrita Vishwa Vidyapeetham (2026) | ₹25,000
- **Third Prize** — VeloHack 2026, Vel Tech University | ₹16,000
- **First Prize** — IDEATHON Hackathon, KGiSL Institute of Technology (2026)
- **Runner-Up** — CRYPTERA 2025, for presenting “LiFi: The Saviour”

Certifications

- **ServiceNow AI Essentials** and **AI Agents & Agentic AI** — ServiceNow (2026)
- **Celonis Foundations** and **Get Data into Celonis** — Celonis Academy (2026)